

Risky Business: Quantitative Portfolio Management for Optimized Risk

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Abstract

We compare portfolio performance using cumulative returns, Sharpe ratio, and the Sortino ratio. Early analysis focuses on whether defensive or diversified portfolios remain more stable during periods of market stress compared to growth-heavy and benchmark portfolios.

The main hypothesis explores whether a Barbell Strategy portfolio can deliver adequate risk adjusted returns, as compared to other sector based portfolios and standardized benchmarks. Risk is measured as Sharpe and Sortino Ratios, and Benchmarks are the S&P 500 and the composite market metric minus the Risk Free Rate (Mkt - Rf).



Introduction & Motivation

Problem & question: This project evaluates how different asset allocations behave across normal market conditions, growth periods, and financial crises. How can portfolio optimization help stakeholders construct portfolios that preserve return potential while reducing drawdowns during periods of financial crisis and market uncertainty?

Impact & stakeholders: This project is designed for stakeholders who need to evaluate portfolio risk beyond simple return. We focus on risk adjusted return for stakeholders that don't need high speed growth, but stable, long-term gains. This is namely institutional funds and risk-averse investors.

Portfolio Makeup: Our **Barbell** is 95% 10-year treasury bills and 5% Volatility Index (VIX) options.

Mega Tech is seven of the largest tech companies (Apple, Microsoft, NVIDIA, Google, Amazon, Meta Broadcom, and Tesla).

Energy is oil and large utilities (ExxonMobil, Chevron, ConocoPhillips, EOG Resources, Schlumberger, NextEra Energy, Duke Energy, Constellation Energy, Southern Company, and Dominion Energy).

Real Estate is the Vanguard REIT ETF, which is a composite of all commercial real estate sectors.

FMCG is consumer staples like groceries and food and beverage (Walmart, Costco, Target, P&G, Unilever, Kimberly-Clark, Coca-Cola, PepsiCo, Nestle, and General Mills).



Background & Related Work

Modern portfolio theory argues that diversification can reduce risk by combining assets with different return patterns. However, crisis periods often challenge diversification because correlations between assets may rise during market stress. Risk-adjusted measures such as the Sharpe ratio and Sortino ratio provide a more complete view of performance than return alone because they account for volatility and downside risk.

This project builds on portfolio theory, backtesting, and crisis-period analysis to test whether portfolio return behavior becomes less predictable during stressful market periods.

The Sharpe Ratio is a standard measure of return/risk. The formula is the average return of a portfolio divided by the standard deviation of the same portfolio. This can be standardized over a period of time, and is more accurate over longer periods.

The Sortino Ratio is the same measurement, but excluding the right tail risk, since the right tail represents positive risk, or the variance associated with gains, as opposed to the variance associated with losses.

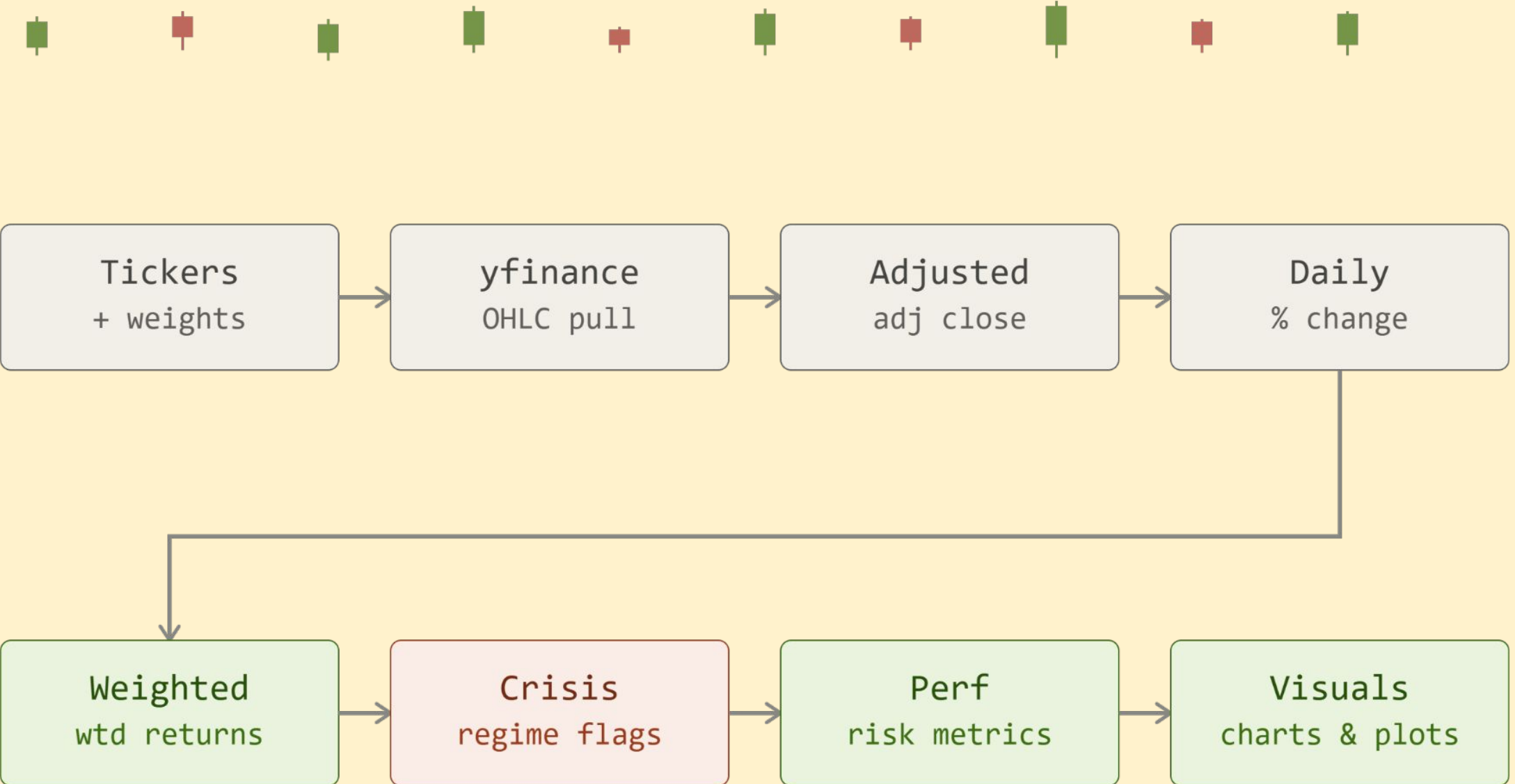


Data, Engineering & Ethics

Sources & scale: The project uses historical daily market data from Yahoo Finance accessed through the **yfinance** Python package. The M2 data scope is frozen through **2026-06-28**. No additional market data will be collected after this milestone.

Pipeline: Ticker list + portfolio weights -> yfinance price pull -> adjusted prices -> daily returns -> weighted portfolio returns -> crisis labels -> performance metrics -> visualizations

Ethics & responsible use: All data used in this project are public financial market data. The project does not use personally identifiable information, private investor records, or restricted financial data.



Methods & Evaluation

Approach: This project combines backtesting, risk analysis, and barbell strategy. Barbell strategy involves putting a large majority of a portfolio in safe assets such as treasury bills, and a small percentage in options that make large gains in times of high volatility.

Validation: Each portfolio is evaluated over the same historical period using daily returns. Portfolio returns are calculated by applying balanced portfolio weights to individual asset returns.

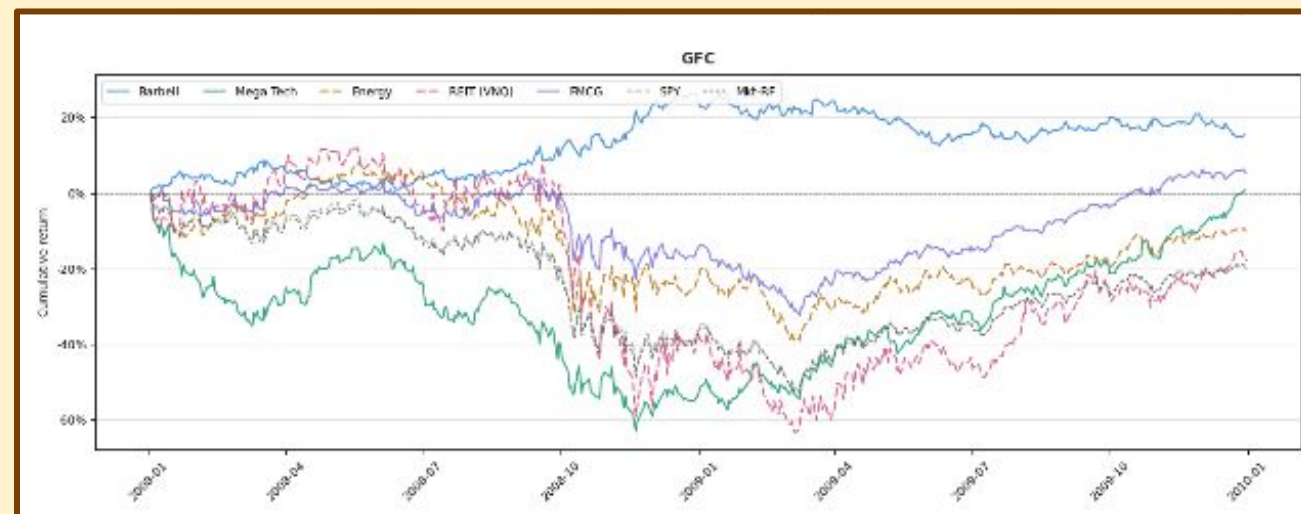
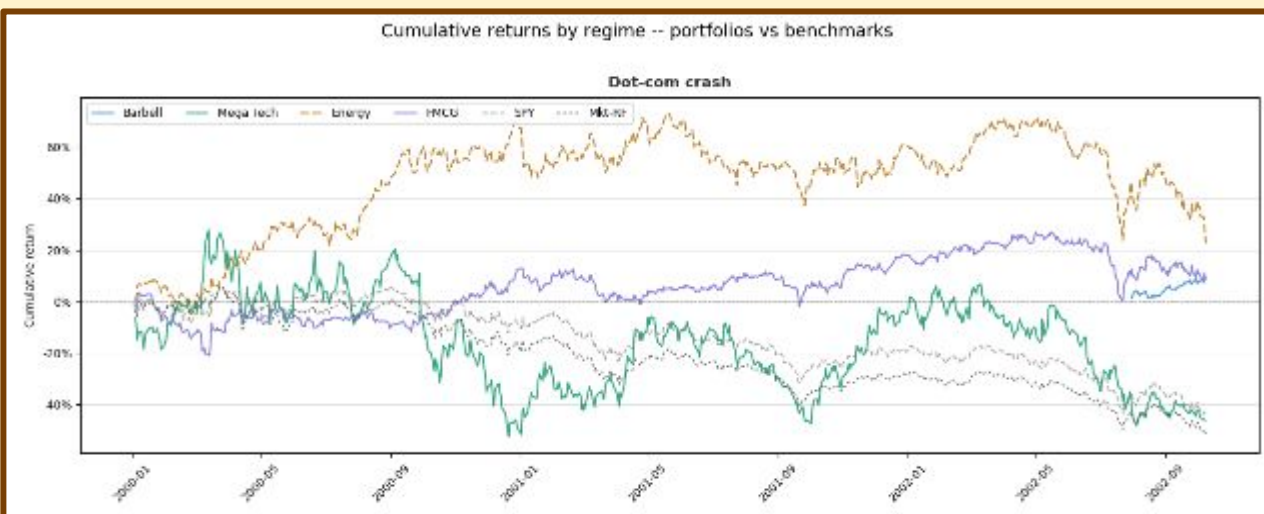
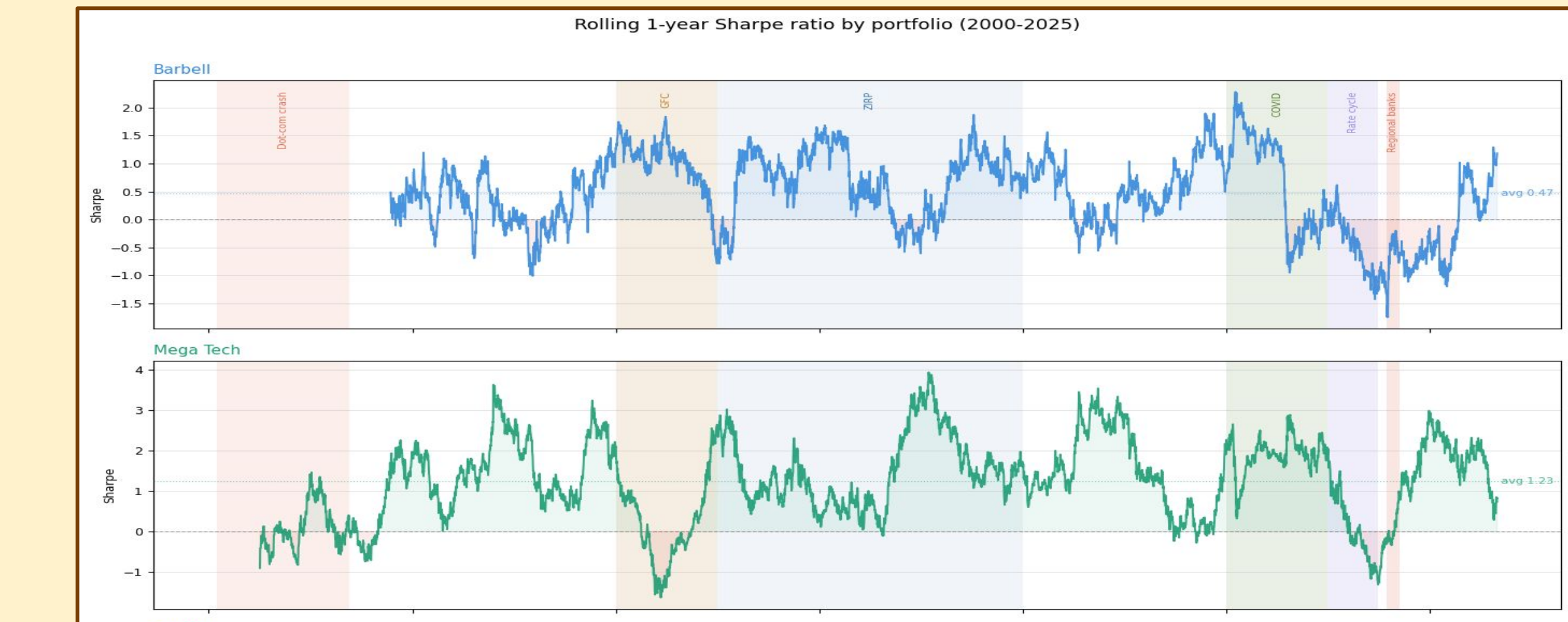
Risk and Performance Metrics

The project evaluates each portfolio using:

- Cumulative return
- Annualized volatility
- Sharpe ratio
- Sortino ratio
- Crisis-period return
- Recovery behavior
- Benchmark comparison



Results & Visualizations



Regime	Sharpe	Sortino	Ann.Ret	Ann.Vol
Barbell	3.720	6.908	45.70%	11.85%
Dot-com crash	0.406	0.664	6.11%	8.67%
Recovery	0.577	1.057	7.88%	12.38%
GFC	0.817	1.584	8.58%	10.42%
Recovery/ZIRP	0.601	1.167	7.98%	9.49%
Pre-COVID	0.802	1.552	8.99%	10.86%
COVID	-1.259	-2.368	-11.15%	10.44%
Rate cycle	0.482	0.930	11.43%	13.64%
Regional banks	0.287	0.507	7.72%	9.76%
Post-crisis				

Figure 1. The rolling sharpe ratio by portfolio for available data between 2000-2025
Figure 2 and 3. Cumulative returns for each drawdown and recovery regime.
Figure 4. Sharpe, Sortino, Annual Return, and Annual Volume for each portfolio
Figure 5. Benchmark performance, S&P 500 and Market minus Risk-Free rate (Mkt - Rf).



Discussion, Limitations & Conclusions

Discussion: The project evaluates portfolio resilience from multiple angles rather than relying on return alone. A tech-based portfolio may generate high cumulative returns but still expose stakeholders to large drawdowns during crises. Similarly, a defensive portfolio may reduce downside risk but underperform during growth periods.

Limitations & future work: Historical performance does not guarantee future performance.

- Backtesting can create false confidence if results are overinterpreted.
- Selected assets may introduce survivorship bias.
- Some assets have shorter histories due to IPO dates or fund creation dates.
- The VIX is an index that tracks volatility options, not a directly investable asset.
- Yahoo Finance data may revise adjusted prices or change ticker availability.
- ARIMA forecasting was used, but failed to reveal any conclusive results.
- This is not financial advice. Just the work of two grad students who spend way too much time tracking markets.

Conclusions: Portfolio resilience should be evaluated using multiple dimensions of performance, including return, volatility, drawdown, risk-adjusted return, and forecast stability. Careful diversification across sector and strategy can help give stable returns and help safeguard against volatility.



Reproducibility & References

Repository: [<https://github.com/awjackson00/data510--Quant-Back-Testing->] · Key dependencies: [numpy, pandas, tidyfinance, seaborn, scikit-learn, yfinance, statsmodel]

References: Markowitz, H. Portfolio Selection. Sharpe, W. Capital Asset Prices and Risk-Adjusted Return. Sortino, F. Downside Risk Framework. Box and Jenkins/ARIMA time-series modeling. Taleb, N. Antifragile: Things That Gain From Disorder. Yahoo Finance and TidyFinance Documentation

